

# VigilVoice

## Voice evidence.

## Verified actions.

A voice-risk detector connected to a guarded approval workflow.

Real pretrained audio inference · Separate verification · Simulated transfers · Auditable decisions

Teacher demonstration

No money moves

# A convincing voice should not be enough to authorize a sensitive action.

## The risk

A synthetic recording can sound persuasive. Poor recordings also make audio assessment unreliable.

The operator needs a clear decision process when evidence is risky or missing.

## Our prototype

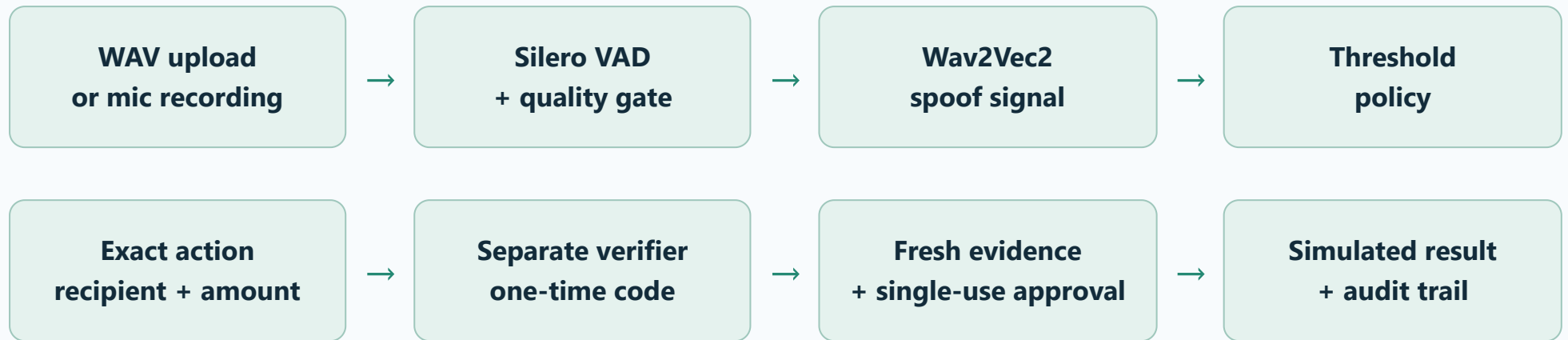
Assess speech quality and spoof signals, then require verification of the exact recipient and amount.

Block completion when evidence is unsafe, missing, stale, or verification fails.

The audio score is a risk signal. It does not establish identity, intent, or a calibrated probability of fraud.

## WHAT IS IMPLEMENTED

# One traceable path from speech to decision.



FastAPI · PostgreSQL · ONNX Runtime on CPU · Native browser UI · Docker

Model weights are from third parties. Our implementation integrates inference with verification, failure handling, and action control. Browser microphone input is record-then-analyze.

# Three cases the audience can follow.

| Input                    | What to demonstrate                                                     | Expected workflow                                                   |
|--------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| Bundled genuine speech   | Analyze → prepare transfer → verifier reviews details → code → complete | LOW/ELEVATED still requires verification                            |
| Bundled synthetic speech | Analyze → prepare another transfer                                      | HIGH blocks completion                                              |
| Your microphone          | Record 3–10 seconds → stop → preview → analyze                          | Show the actual output, including false alarms or quality rejection |

A second tab simulates delivery; it is not an independent authentication factor. Every transfer is simulated. Keep the verifier key private.

Rehearsal: file evidence 120s · code 60s · approval 60s · maximum 3 incorrect code attempts. Expired/blocked action? Start a new transfer.

# Working controls, exploratory ML results.

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## Backend check groups passed

Action binding, session isolation, concurrent completion, code limits/expiry, stale evidence, unsafe audio, and disconnect handling.

Real running API and models; expiry tests age only their own test records.

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## Public labeled audio clips

20 validation + 20 test clips. Source-recording groups separated; bundled demo sources excluded.

Exploratory experiment. Training-data overlap, speaker overlap, and target-language coverage are unverified.

| 20-clip test partition                    | Default HIGH 0.75 | Offline candidate 0.58 |
|-------------------------------------------|-------------------|------------------------|
| Synthetic clips eligible for verification | 1 / 9 scorable    | 0 / 9 scorable         |
| Genuine clips falsely blocked as HIGH     | 1 / 9 scorable    | 3 / 9 scorable         |
| Quality rejections (excluded above)       | 2 / 20 total      | 2 / 20 total           |

Candidate chosen on validation only; remains offline. Lowering the threshold caught more synthetic clips but blocked more genuine speech. Median offline file inference: 3.06s; p95: 7.79s.

Full results and exact denominators: [evaluation-report.md](#). Software checks demonstrate those scenarios, not production security certification.

# What we can—and cannot—claim today.

| Implemented and demonstrated                                                                                                                | Still pending                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Real pretrained VAD and spoof inference                                                                                                     | Our own model training or fine-tuning                               |
| Public-corpus experiment with held-out test partition                                                                                       | Representative Indian-language, replay and phone-channel validation |
| Offline threshold candidate evaluation                                                                                                      | Production threshold/probability calibration                        |
| Local verifier and transaction simulation                                                                                                   | Trusted delivery and real payment integration                       |
| Microphone recording and WAV input                                                                                                          | Continuous browser streaming and field/load trials                  |
| Present this as an engineering prototype. Do not claim an accuracy percentage for real-world use or that all SIH PDF promises are complete. |                                                                     |

## NEXT DEVELOPMENT STEPS

# Make the evidence stronger, then expand deployment.

1. Collect consented Indian-language, noisy phone, replay, and diverse synthetic speech.
2. Use separated source/speaker splits to evaluate and calibrate; fine-tune only if the results justify it.
3. Add trusted verification delivery, accounts, HTTPS, and explicit audio retention.
4. Measure concurrency and stream latency before a wider pilot.

Today's deliverable: a reproducible local demonstration, test receipts, and an honest evaluation report.